Project: Worked examples
Title: Insulation thickness (ISO 12241 formula route)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The insulation thickness on a pipe (cylinder) or a flat surface that meets a heat-flow limit, a limit on the surface temperature of a hot surface and an anti-condensation requirement for a cold one, each solved separately and the governing one reported; and for a chosen thickness the heat flow, the surface temperature and PASS/FAIL per criterion. Steady state, one layer, no thickness table of any standard. Pipe resistance per metre: R = ln(D_e/D_i)/(2π·λ) + 1/(h_se·π·D_e), D_e = D_i + 2t; with the inner film and pipe wall R_oth = 1/(h_i·π·D_p) + ln(D_i/D_p)/(2π·λ_w) added in series Flat resistance per square metre: R = t/λ + 1/h_se (+ 1/h_i + t_w/λ_w) Heat flow: q = (θ_i − θ_a)/R, positive from the process to the ambient Surface temperature: θ_se = θ_a + q/(h_se·π·D_e) (pipe), θ_a + q/h_se (flat) Surface coefficient: h_se as given, or h_se = h_c + h_r with h_r = ε·σ·(T_s² + T_a²)(T_s + T_a); h_c from free convection to air (laminar or turbulent, the larger), forced convection (cross-flow cylinder or flat plate) and their cube-law blend; solved with θ_se, which it depends on Heat-loss limit: the thickness beyond which |q| ≤ q_max; surface-temperature limit (hot): θ_se ≤ θ_se,max; anti-condensation (cold): θ_se ≥ θ_dew + margin, θ_dew entered (for example from B2) or from the Magnus relation at θ_a and RH Solve: each criterion's thickness by scanning t from 0 to 1 m for the last broken point and bisecting the bracket after it; governing t = max over the selected criteria Critical diameter: D_crit = 2λ/h_se; below it, insulation raises the heat flow until D_e passes D_crit

Ref: ISO 12241 formula route (clause numbers to be checked) — Thermal insulation for building equipment and industrial installations: calculation rules
Ref: Incropera, Fundamentals of Heat and Mass Transfer 7th ed., §7.2, §7.4, §9.6, §9.9 (to be checked) — External forced, free and mixed convection correlations
Ref: ISO 13788 cl. 6 (clause number to be checked against the owner's copy) — Magnus saturation vapour pressure (shared with B7)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
geometry	Surface	pipe (cylinder)		assumption: A round pipe.	any
D_i	Pipe outer diameter, the inside of the insulation (pipe)	90	mm	drawing: M-PIP-014 rev A	10 ... 1000 mm
theta_i	Process (fluid) temperature	6	degC	spec: SPEC-M-11 §3: chilled-water flow temperature	-50 ... 400 degC
theta_a	Ambient air temperature (also the radiation sink)	28	degC	spec: SPEC-M-11 §2: plant-room design temperature	-20 ... 50 degC
lambda	Thermal conductivity of the insulation	0.035	W/(m*K)	assumption: Closed-cell elastomeric insulation; take the maker's value at the mean temperature (W3 material)	0.015 ... 0.2 W/(m*K)

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h_mode	Where the outer surface coefficient comes from	h_se as given		assumption: A surface coefficient is given, not computed.	any
h_se	Outer surface coefficient h_se (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)	8	W/(m^2*K)	assumption: Still air in a plant room; replace with the project's own record.	3 ... 40 W/(m^2*K)
eps	Emissivity of the outer surface (for the correlation; with h_se as given it only judges that value against condensation on a cold surface)	0.9		assumption: Painted or plastic jacket (invented for this example); with h_se given it only judges that value against condensation.	0.05 ... 1
v_wind	Air speed over the surface (correlation; 0 = still air)	0	m/s	assumption: Not used: h_se is given.	0 ... 20 m/s
L_surf	Height or flow length of the flat surface (correlation, flat)	1	m	assumption: Not used: a pipe.	0.1 ... 10 m
q_max_m	Heat-loss limit per metre of pipe (generic default of 25 W/m, replace with your project limit; blank = not a criterion)	12	W/m	spec: SPEC-M-11 §5: heat gain not to exceed 12 W/m	1 ... 1000 W/m
dew_margin	Margin of the surface above the dew point, for a cold surface (blank = not a criterion)	1	K	spec: SPEC-M-11 §5: surface at least 1 K above the dew point	0 ... 5 K
RH_a	Relative humidity of the ambient air (anti-condensation)	70	%	spec: SPEC-M-11 §2: plant-room design humidity	20 ... 95 %
t_chosen	Chosen insulation thickness (blank = report at the required thickness, no checks; with it, nothing blocks)	25	mm	spec: Pipework schedule M-SCH-014: 25 mm sleeves	0 ... 300 mm

3. Assumptions

A1. Steady one-dimensional heat flow through a single homogeneous layer of constant conductivity; no thermal bridges at supports, no joints, no moisture in the insulation. λ is taken at the value entered: use the value at the mean insulation temperature shown on the sheet (optionally noted in 'Temperature the conductivity applies at'). (Formula route of the standard.)
A2. With the correlation, h_se = h_c + h_r. Radiation: h_r = ε·σ·(T_s² + T_a²)(T_s + T_a) to surroundings at the ambient air temperature. Free convection to air: Holman's simplified relations at atmospheric pressure, the larger of the laminar and turbulent forms (horizontal cylinder 1.32(ΔT/D)^0.25 and 1.24ΔT^(1/3); vertical

surface 1.42(ΔT/L)^0.25 and 1.31ΔT^(1/3)). Forced convection: Churchill–Bernstein for a cylinder in cross-flow, the laminar/mixed flat-plate relations along a plate; air properties from fits at the film temperature, Pr = 0.71; blended with free convection as (h_nat³ + h_forced³)^(1/3). Indicative, not the standard's own correlation. (Incropera 7th ed. §7.2, §7.4, §9.6 free convection and §9.9 mixed convection (section numbers to be checked against the owner's copy); the free-convection constants are Holman's, Heat Transfer; the standard's own coefficients are not reproduced.)
A3. The dew point is entered (for example from B2) or is the Magnus relation at θ_a and RH, the same closed form B7 uses. Below 0 °C the relation is over ice, so RH is then taken relative to ice, whereas meteorological RH is usually relative to water: enter the dew point for sub-zero air. A margin above it is the user's. (Derived; B7 shares the function.)
A4. Pipes are treated as horizontal for free convection and flat surfaces as vertical; air properties are fits (conductivity within about 3 %, viscosity within about 4 % over 250–400 K), Pr = 0.71. (Stated.)
A5. The required thickness is the least thickness beyond which the criterion holds for every thicker layer up to 1 m, which is what matters when thin insulation on a small pipe raises the heat flow (critical diameter). (Stated.)
A6. No thickness table is used or shipped: not BS 5422, not the tables of any standard. The conductivity is the user's input, from a W3 material or the maker's data. (No standard's table is reproduced.)
A7. geometry (Surface) is taken as pipe (cylinder). (A round pipe.)
A8. lambda (Thermal conductivity of the insulation) is taken as 0.035 W/(m*K). (Closed-cell elastomeric insulation; take the maker's value at the mean temperature (W3 material))
A9. h_mode (Where the outer surface coefficient comes from) is taken as h_se as given. (A surface coefficient is given, not computed.)
A10. h_se (Outer surface coefficient h_se (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)) is taken as 8 W/(m^2*K). (Still air in a plant room; replace with the project's own record.)
A11. eps (Emissivity of the outer surface (for the correlation; with h_se as given it only judges that value against condensation on a cold surface)) is taken as 0.9. (Painted or plastic jacket (invented for this example); with h_se given it only judges that value against condensation.)
A12. v_wind (Air speed over the surface (correlation; 0 = still air)) is taken as 0 m/s. (Not used: h_se is given.)
A13. L_surf (Height or flow length of the flat surface (correlation, flat)) is taken as 1 m. (Not used: a pipe.)

4. Working

[11] theta_dew = inverse Magnus of RH·p_sat(□_a) — ISO 13788 (Magnus form, as B7)
= □_a = 28 °C, RH = 70 %
= 22.012 degC
Dew point of the ambient air

[12] t_q = t such that |q| ≤ 12 W/m for every thicker layer — ISO 12241
= scan 0 … 1000 mm, then bisection
= 17.814 mm
Required thickness, heat-gain limit

[13] t_dew = t such that □_se ≥ 23.012 °C for every thicker layer — ISO 12241
= scan 0 … 1000 mm, then bisection
= 13.162 mm
Required thickness, anti-condensation margin

[14] thickness = max over the selected criteria — ISO 12241
= t_q = 17.814 mm, t_dew = 13.162 mm
= 17.814 mm
Governing required thickness (heat-gain limit)

[16] D_e = D_i + 2t
= 90 mm + 2 × 25 mm
= 140 mm
Outer diameter of the insulation, chosen thickness

[17] R_ins = ln(D_e/D_i)/(2π·λ) ISO 12241
= ln(140/90)/(2π × 0.035)
= 2.0091 m·K/W
Resistance of the insulation
[18] R_se = 1/(h_se·π·D_e) ISO 12241
= 1/(8 × π × 0.14 m)
= 0.28421 m·K/W
Resistance of the outer surface
[19] R = R_oth + R_ins + R_se ISO 12241
= 0 + 2.0091 + 0.28421
= 2.2933 m·K/W
Total resistance per metre
[20] q_per_m = |θ_i - θ_a| / R ISO 12241
= 22 K / 2.2933
= 9.593 W/m
Heat flow per metre of pipe (heat gain, magnitude)
[21] theta_se = θ_a + q/(h_se·π·D_e) ISO 12241
= 28 °C - 9.593 / 3.5186
= 25.274 degC
Outer surface temperature
[22] theta_m = (θ_i + θ_se)/2
= (6 + 25.274)/2
= 15.637 degC
Mean temperature of the insulation: take λ at this temperature
[23] D_crit = 2λ/h_se at the bare surface ISO 12241
= 2 × 0.035 / 8
= 8.75 mm
Critical diameter at the bare-surface h_se: below it, insulation raises the heat flow
[24] u_q = |q| / q_max
= |q| ≤ 12 W/m
= 0.79941
Utilisation, heat-gain limit
[25] u_dew = (θ_a - θ_se)/(θ_a - (θ_dew + margin))
= θ_se ≥ 23.012 °C
= 0.54661
Utilisation, anti-condensation margin

5. Results

SYMBOL	VALUE	DESCRIPTION
theta_dew	22.012 degC	Dew point of the ambient air
t_q	17.814 mm	Required thickness, heat-gain limit
u_q	0.79941	Utilisation, heat-gain limit
t_dew	13.162 mm	Required thickness, anti-condensation margin
u_dew	0.54661	Utilisation, anti-condensation margin
thickness	17.814 mm	Governing required thickness
q_per_m	9.593 W/m	Heat flow per metre (heat gain)
theta_se	25.274 degC	Outer surface temperature
theta_m	15.637 degC	Mean insulation temperature
R	2.2933 m·K/W	Total resistance

SYMBOL	VALUE	DESCRIPTION
D_crit	8.75 mm	Critical diameter (at the bare-surface h_se)
governing	"heat-gain limit"	Governing criterion

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Chosen thickness against the heat-gain limit	u_q = 0.79941	u_q ≤ 1	0.799	ISO 12241
PASS	Chosen thickness against the anti-condensation margin	u_dew = 0.54661	u_dew ≤ 1	0.547	ISO 12241

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: lambda (Material properties); h_se (Outer surface coefficient); eps (Surface emissivity); q_max_m, dew_margin (Insulation criteria). Enter each as one so its source, edition and licence travel with it.

8. Verification

Template bs.insulation@1.1.0 Content hash 027d4f4ba606eddd Engine calc-engine/1.0.0
Known-answer cases passed 14/14 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0